

Notes: Brown, Kowalski & Lurie (RES, 2020)

Long-Term Impacts of Childhood Medicaid Expansions on Outcomes in Adulthood

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1 Big picture

- **Question.** What are the long-run effects of expanded childhood Medicaid eligibility on adult education, fertility, mortality, earnings, tax transfers, tax payments, and government fiscal returns?
- **Policy setting.** Federal and state Medicaid/SCHIP policy changes in the 1980s and 1990s created variation in children’s Medicaid eligibility across states, birth months, birth years, ages, and household poverty status.
- **Data advantage.** The paper uses IRS administrative tax data linked to Social Security Administration information. This permits population-scale observation of adult outcomes from ages 19 to 28 for children born from 1981 through 1984.
- **Treatment measure.** The key treatment is *simulated years eligible for Medicaid from ages 0 to 18*. The authors use a Medicaid eligibility calculator based on state/federal policy rules and apply it to a fixed national simulation sample.
- **Main empirical design.** The baseline reduced form regresses adult outcomes at each age on simulated Medicaid eligibility, controlling for birth-month cohort fixed effects and state-of-residence-at-age-15 fixed effects.
- **Main results.** More childhood Medicaid eligibility increases college enrollment, delays fertility, lowers adult mortality, raises female wage income, lowers EITC receipts, and raises total taxes paid.
- **Fiscal result.** By age 28, each additional year of simulated childhood Medicaid eligibility raises future taxes enough that the government recoups a large share of spending; at a 3% discount rate it recoups 58 cents per dollar spent.
- **Distributional result.** Effects are largest for individuals from low-FPL households, which supports the interpretation that the Medicaid channel, rather than state-by-cohort confounds affecting all children, drives the estimates.
- **Economic takeaway.** Childhood Medicaid is not only a health-insurance program. It is also a long-run human capital, fertility-timing, mortality, and public-finance intervention.

1.1 Incremental contribution of the paper

- **Moves beyond short-run Medicaid outcomes.** Earlier Medicaid work documented insurance coverage, utilization, and near-term health effects. This paper traces children into adulthood and estimates outcomes year by year from ages 19 to 28.
- **Uses population-scale administrative data.** The IRS data give a massive sample, very low attrition for administrative outcomes, and measures of education, fertility, wages, EITC, and taxes that are hard to observe together in survey data.
- **Links health policy to tax administration.** The paper connects childhood Medicaid eligibility to adult tax receipts, allowing a fiscal-return calculation rather than only a social-benefit narrative.
- **Shows age profiles rather than single endpoints.** The figures trace timing: college and fertility effects appear early, wage and tax effects emerge later, and cumulative fiscal impacts grow with age.
- **Develops and distributes an eligibility calculator.** The calculator codifies federal and state Medicaid thresholds for children born January 1981 to December 1984, making the simulated eligibility design transparent and replicable.
- **Provides a dose-response validation.** By comparing low-, middle-, and high-FPL childhood households, the paper tests whether effects are strongest where Medicaid eligibility should matter most.
- **Quantifies partial fiscal payback.** The paper shows that long-run tax revenue offsets a substantial fraction of childhood Medicaid spending by age 28, even before adding private welfare gains and the value of reduced mortality.

2 Data and measurement

2.1 Administrative tax data and birth cohorts

- The main data come from administrative tax records from the IRS, beginning in 1996.
- The sample consists of children born from January 1981 through December 1984.
- These cohorts are old enough for adult outcomes through age 28 and young enough to be linked to parents in 1997, the first year in which the authors are confident in the parent-child linkage.
- The baseline requires the child to be claimed on a 1997 Form 1040 and requires parents to file in each year from 1996 through the year in which the child turns 18.
- After de-duplication, state-residence requirements, and parental filing restrictions, the main estimation sample contains 10,045,162 children: 4,913,139 females and 5,132,023 males.

Interpretation: The sample restriction improves the accuracy of childhood residence and income/FPL measures. The cost is that some children whose parents do not file in every year are excluded; the paper later shows robustness using an expanded sample with imputed state and income information.

2.2 Medicaid eligibility calculator

- The tax data do not contain actual Medicaid eligibility or enrollment for every child.
- The authors construct a Medicaid eligibility calculator using federal and state Medicaid rules.
- Inputs are birth month and year, state of residence, household income as a percent of the Federal Poverty Level (FPL), household size, year, and age.
- Eligibility is calculated monthly, but the paper mainly uses December eligibility because the tax data provide annual household information.
- Actual years eligible from birth to age 18 average 3.77 years, while simulated years eligible average 4.49 years.

Interpretation: The calculator translates complex state and federal Medicaid laws into a standardized eligibility measure. Because eligibility rules depend on both age and birth month, children born in different months can face different eligibility histories even within the same state.

2.3 Simulated eligibility instrument

- To isolate policy-driven eligibility variation, the paper constructs simulated eligibility in the tradition of Currie and Gruber.
- The authors draw a national simulation sample of 200,000 dependents from 1997.
- For each state, calendar year, and birth month, the calculator computes the share of the simulation sample eligible for Medicaid.
- Each child in the main tax sample is assigned simulated eligibility according to state of residence and birth-month cohort.
- The baseline treatment variable is cumulative simulated eligibility from ages 0 to 18:

$$Z_i^{0-18} = \sum_{t=0}^{18} Z_{i,t}. \quad (1)$$

Interpretation: Simulated eligibility removes direct dependence on the child’s own household income and thus reduces endogeneity from family economic circumstances. It preserves variation generated by policy rules that treat similar children differently across states, birth months, and ages.

2.4 Adult outcomes

- **College enrollment.** Measured from Form 1098-T filed by educational institutions. Outcomes include current enrollment at age a and ever enrolled by age a .
- **Fertility.** Measured from whether an individual ever claims a dependent child on Form 1040. SSA dates of birth for dependents allow the parent age at first dependent-child birth to be inferred.
- **Mortality.** Measured from SSA death records linked to tax data.
- **Wage income.** Measured from employer-filed Form W-2, so wages are observed even for many non-filers.
- **EITC receipts.** Measured from tax returns. Non-filers are assigned zero EITC.
- **Total taxes.** Defined as household federal tax payments plus individual payroll tax payments less EITC, adjusted to 2011 dollars.

Interpretation: The outcomes cover mechanisms and fiscal consequences. College and fertility measure human capital and family timing; mortality measures health; wages, EITC, and taxes measure labor-market and fiscal effects.

2.5 Contemporaneous versus cumulative outcomes

For most outcomes, the paper estimates two versions:

- **Contemporaneous outcomes:** outcome at a specific adult age a .
- **Cumulative outcomes:** outcome accumulated from age 19 through age a .

Interpretation: Contemporaneous outcomes show timing. Cumulative outcomes show aggregate effects by a given age and are more relevant for fiscal-return calculations.

2.6 Medicaid takeup and spending data

- To convert eligibility into enrollment and spending, the paper uses Medicaid Statistical Information System (MSIS) data.
- MSIS data are aggregated by state and year for children under 21, so the authors disaggregate by age and cohort using eligibility shares from the calculator and CPS-based population estimates.
- Spending is adjusted to 2011 dollars.
- Takeup and spending outcomes are assigned to children in the tax data in the same way adult outcomes are assigned.

Interpretation: The fiscal-return calculation requires spending, not just eligibility. Estimating takeup and spending with the same simulated eligibility variation avoids importing takeup rates from other studies that may be based on different complier populations.

3 Models and empirical specifications

3.1 Baseline reduced-form specification

For adult outcome $Y_{i,a}$ at age a , the paper estimates:

$$Y_{i,a} = \beta_a \sum_{t=0}^{18} Z_{i,t} + \gamma_c + \gamma_s + \varepsilon_{i,a}. \quad (2)$$

- $Y_{i,a}$ is an adult outcome measured at age a .
- $Z_{i,t}$ is simulated Medicaid eligibility at childhood age t .
- γ_c are birth-month cohort fixed effects.
- γ_s are state-of-residence-at-age-15 fixed effects.
- β_a is the effect of one additional simulated year of childhood Medicaid eligibility.

Interpretation: The coefficient is a reduced-form effect of policy-induced eligibility exposure, not directly an effect of Medicaid enrollment. The paper prefers this transparent reduced form because the policy lever is eligibility and because the first stage is large.

3.2 Age-by-age outcome profiles

$$\{\hat{\beta}_{19}, \hat{\beta}_{20}, \dots, \hat{\beta}_{28}\} \quad (3)$$

are estimated from separate regressions for each adult age.

Interpretation: The age profiles are central to the paper. They reveal that effects on college and fertility appear early, wage and tax effects appear later, and fiscal effects accumulate gradually.

3.3 Gender-specific estimates

The same specification is estimated in:

- the full sample,
- the female sample,
- the male sample.

Interpretation: Gender differences are important because fertility, wage growth, EITC eligibility, and mortality can respond differently for women and men. The paper generally finds stronger wage effects for women and similar or imprecise effects for men.

3.4 Dose-response by childhood FPL

The paper estimates the reduced form separately for three childhood household-FPL groups:

- high-impact group: below 200% of FPL in every observed childhood year,
- intermediate-impact group: 200%-500% FPL,
- low-impact group: above 500% FPL.

Interpretation: Medicaid eligibility should matter most for poorer children. A dose-response pattern reduces concern that the estimates are driven by state-cohort shocks affecting all children equally.

3.5 Fiscal return on investment

The fiscal comparison uses reduced-form coefficients for spending and taxes. For discount rate r :

$$\text{Fiscal ROI}_r = \frac{\hat{\beta}_r^{\text{taxes}}}{\hat{\beta}_r^{\text{spending}}} - 1. \quad (4)$$

At a 3% discount rate, Table 1 reports $\hat{\beta}^{\text{spending}} = 404$ and $\hat{\beta}^{\text{taxes}} = 233$, so:

$$\frac{233}{404} \approx 0.58, \quad \text{Fiscal ROI}_{3\%} \approx -42\%. \quad (5)$$

Interpretation: The federal/state government does not fully recoup spending by age 28 through taxes alone. But the payback is large and incomplete by construction: the adults are only age 28, and the calculation excludes private welfare gains and the value of reduced mortality.

3.6 Takeup-scaled interpretation

Because one simulated year of eligibility increases Medicaid takeup by about 0.59 years, an approximate enrollment effect can be computed as:

$$\text{Effect per year enrolled} \approx \frac{\text{Reduced-form eligibility effect}}{0.59}. \quad (6)$$

Interpretation: This scaling is useful but more assumption-laden than the reduced form. It assumes impacts are zero for those who do not enroll and uses the same variation to estimate takeup.

4 Identification strategy

4.1 Core identifying variation

The design uses variation in simulated Medicaid eligibility across:

1. children born in different birth-month cohorts within the same state,
2. children born in the same birth-month cohort across different states,
3. age-specific policy rules that make eligibility depend on both age and month of birth.

Interpretation: The strategy compares children exposed to different Medicaid eligibility histories because of policy timing and state rules, while holding constant broad state differences and national cohort patterns.

4.2 Role of fixed effects

- Birth-month cohort fixed effects absorb national cohort shocks, such as macroeconomic conditions affecting all children born in the same month.
- State fixed effects absorb time-invariant differences across states, such as persistent health, education, or income differences.

- Clustering standard errors by state allows arbitrary correlation within states over time.

Interpretation: The design does not compare rich and poor children directly. It compares policy-induced eligibility intensity after removing fixed state and cohort components.

4.3 Why simulated eligibility helps

- Actual eligibility depends on household income, which can be correlated with unobserved family resources.
- Simulated eligibility applies policy rules to a fixed national sample, isolating variation from the rules rather than from each family's income.
- This is analogous to a shift-share or simulated-instrument design.

Interpretation: Simulated eligibility is not perfectly random, but it is designed to be more plausibly exogenous than actual eligibility.

4.4 Dose-response validation

- Medicaid eligibility is most relevant for low-income children.
- If state-by-cohort factors unrelated to Medicaid drove the estimates, effects would likely appear similarly across FPL groups.
- Figure 8 shows the largest cumulative tax effects in the low-FPL high-impact sample.

Interpretation: Dose-response is a key credibility test. It shows that the outcomes respond most where Medicaid should bind most strongly.

4.5 Why the reduced form is preferred over IV

The paper emphasizes the reduced form because:

- eligibility thresholds are the policy lever;
- the first stage is close to one for actual eligibility and sizable for takeup;
- reduced-form dose-response patterns are interpretable for children far from poverty, whereas IV estimates can be unstable when the first stage is near zero;
- IV would require more household-FPL information and stronger assumptions about actual eligibility measurement.

Interpretation: The reduced form answers a policy question: what is the impact of increasing eligibility exposure? It does not require treating enrollment as the sole causal treatment.

4.6 Robustness and remaining threats

- **OBRA 90 discontinuity.** A supplemental regression-discontinuity exercise exploits a federal cutoff by birth month, but it is noisier than the preferred design.
- **Expanded sample.** The authors impute state and income for parental non-filers and obtain similar cumulative tax results.

- **OLS and income controls.** OLS is negative because Medicaid targets poor families; reduced-form estimates remain positive with or without income controls, though income controls attenuate estimates.
- **State-specific trends.** Adding state-specific linear trends slightly raises magnitudes but lowers precision.
- **Sibling/family controls.** Sibling and family fixed-effect style exercises attenuate estimates, consistent with limited within-family variation and possible household spillovers.

Interpretation: The remaining threats are state-by-cohort shocks that differentially affect poor children and align with Medicaid expansions. The paper’s strongest defenses are simulated policy variation, fixed effects, dose response, and robustness checks.

5 Tables and figures

5.1 Figure 1: State Variation in Simulated Years Eligible for Medicaid, Ages 0–18

Read:

- The figure maps simulated years of Medicaid eligibility for children born in January 1981 and December 1984.
- The younger cohort has more eligibility in most states, reflecting Medicaid/SCHIP expansions.
- There remains substantial cross-state variation within each cohort.

Interpretation: The map visualizes the treatment variation. The empirical design uses this variation plus birth-month variation and within-childhood age rules.

Economic message: Medicaid eligibility expanded unevenly across the United States. This unevenness creates quasi-experimental variation in childhood public insurance exposure.

5.2 Figure 2: Contemporaneous and Cumulative College Enrollment (%)

Read:

- Contemporaneous college enrollment effects are largest at ages 19–22.
- At age 19, an additional simulated year of Medicaid eligibility raises college enrollment by about 1.69 percentage points in the full sample.
- Cumulative college enrollment remains positive at later ages, but statistical significance fades after about age 26.
- Female and male point estimates are similar despite different baseline enrollment rates.

Interpretation: Medicaid appears to accelerate college enrollment and may increase overall enrollment. The timing overlaps with the college-going years, suggesting a human capital channel.

Economic message: Childhood health insurance can affect education decisions long after childhood, potentially by improving health, family resources, or academic preparation.

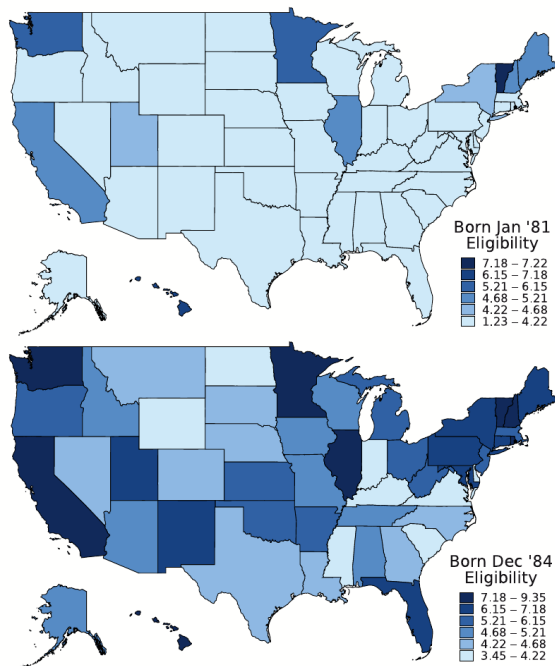


FIGURE 1
State Variation in Simulated Years Eligible for Medicaid, Ages 0–18
Note. Bins reflect the sextiles of the distribution for the cohort born in December 1984. We present the January 1981 and December 1984 cohorts because they are the oldest and youngest cohorts in our sample.

Figure 1: State Variation in Simulated Years Eligible for Medicaid, Ages 0–18

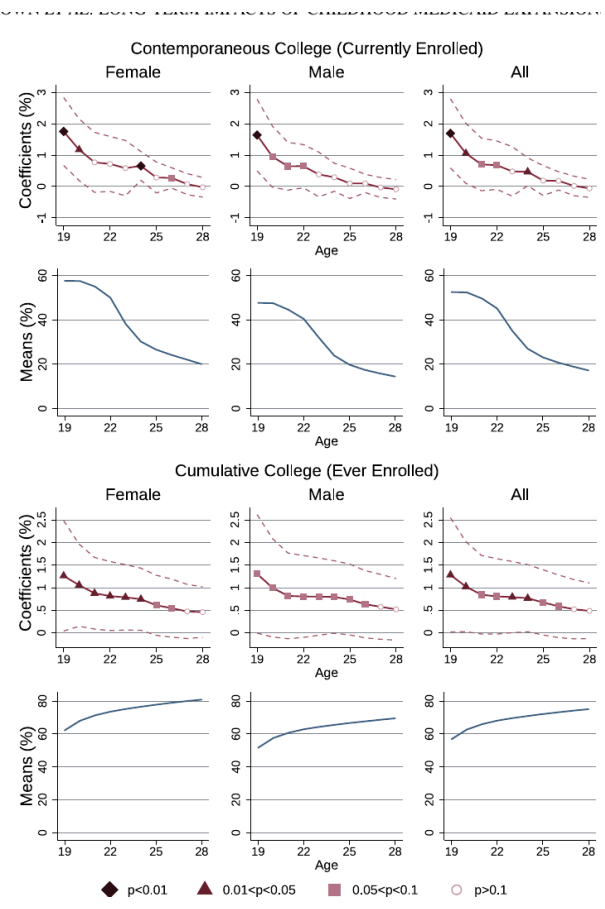


Figure 2: Contemporaneous and Cumulative College Enrollment (%)

5.3 Figure 3: Contemporaneous and Cumulative Fertility (%)

Read:

- Medicaid eligibility reduces early fertility, especially from ages 18 to 21.
- By age 19, each additional year of eligibility lowers the probability of first dependent-child birth by about 0.35 percentage points.
- Cumulative effects shrink as delayed births occur later, but a negative effect remains by age 28.
- Female means are higher than male means, but coefficients are not sharply different across genders.

Interpretation: The fertility results are consistent with Medicaid shifting human capital and family formation decisions. The overlap with college enrollment effects suggests delayed fertility could be one channel through which Medicaid affects later economic outcomes.

Economic message: Medicaid exposure may reduce constraints in adolescence and early adulthood, leading to delayed childbearing and more investment in schooling.

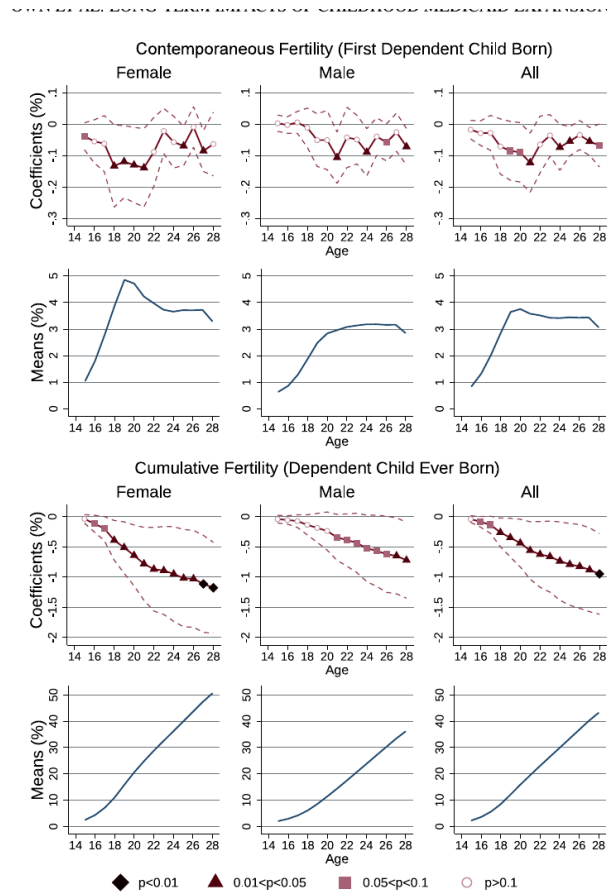


Figure 3: Contemporaneous and Cumulative Fertility (%)

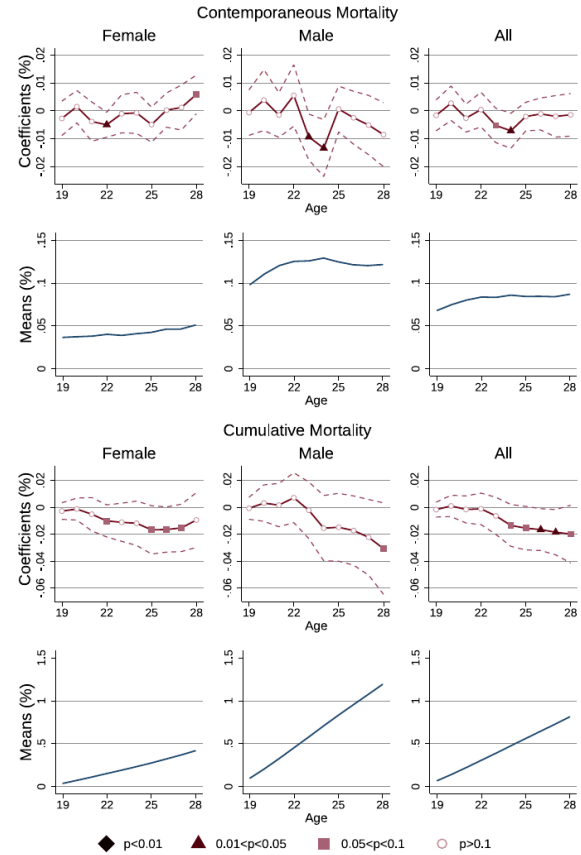


Figure 4: Contemporaneous and Cumulative Mortality (%)

5.4 Figure 4: Contemporaneous and Cumulative Mortality (%)

Read:

- Contemporaneous mortality effects are noisy because mortality rates are low.
- Cumulative mortality by age 28 falls for individuals with more childhood Medicaid eligibility.
- The full-sample cumulative effect by age 28 is roughly a 0.020 percentage point reduction, or about 2 lives per 10,000.

Interpretation: The mortality result is economically important even though the percentage-point effect is small. Mortality is rare at these ages, so small coefficient changes can imply meaningful welfare gains.

Economic message: Medicaid's benefits extend beyond fiscal tax receipts: lower mortality has direct welfare value and strengthens the social return calculation.

5.5 Figure 5: Contemporaneous and Cumulative Wage Income (\$000)

Read:

- Average wage income rises strongly with age.
- Wage effects become visible around age 23.
- Female wage effects are positive and grow with age; male estimates are smaller and less precise.
- By age 28, an additional year of simulated eligibility raises cumulative wage income by about \$1,784 for females.

Interpretation: The wage results are consistent with education and fertility channels. They also show why fiscal effects grow later: labor-market returns appear after early adulthood transitions.

Economic message: Medicaid's effect on earnings is delayed. Evaluating only early adult outcomes would understate long-run fiscal and labor-market gains.

5.6 Figure 6: Contemporaneous and Cumulative Earned Income Tax Credit (EITC, \$000)

Read:

- Medicaid exposure reduces EITC receipts at most adult ages.
- Cumulatively by age 28, each additional simulated year of eligibility lowers EITC receipts by about \$182 on a base of \$3,044.
- Effects are similar across genders in the full set of ages, although the timing differs.

Interpretation: Lower EITC receipts can reflect higher earnings, delayed fertility, or changes in filing status and family structure. It is a fiscal gain to the government but should not be interpreted mechanically as a welfare loss to recipients.

Economic message: A social program can affect the cost of other transfer programs through later earnings and family choices.

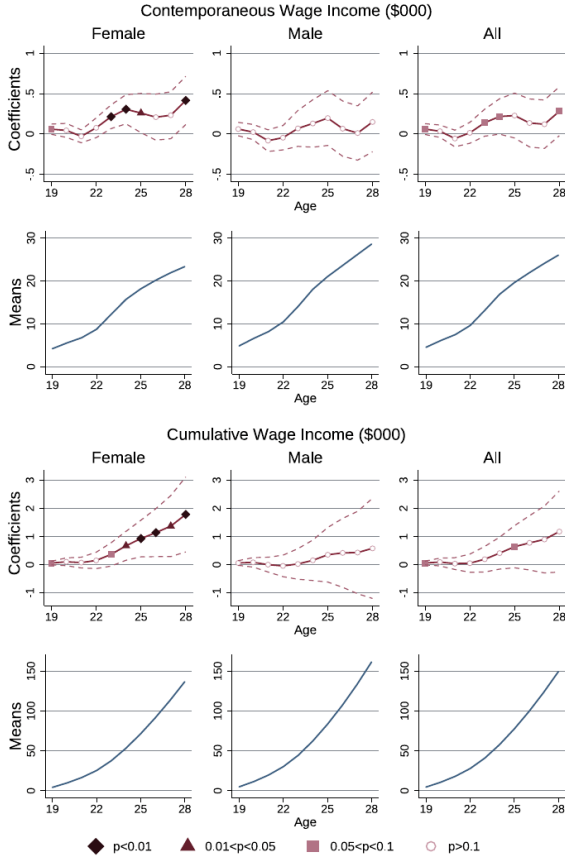


Figure 5: Contemporaneous and Cumulative Wage Income (\$000)

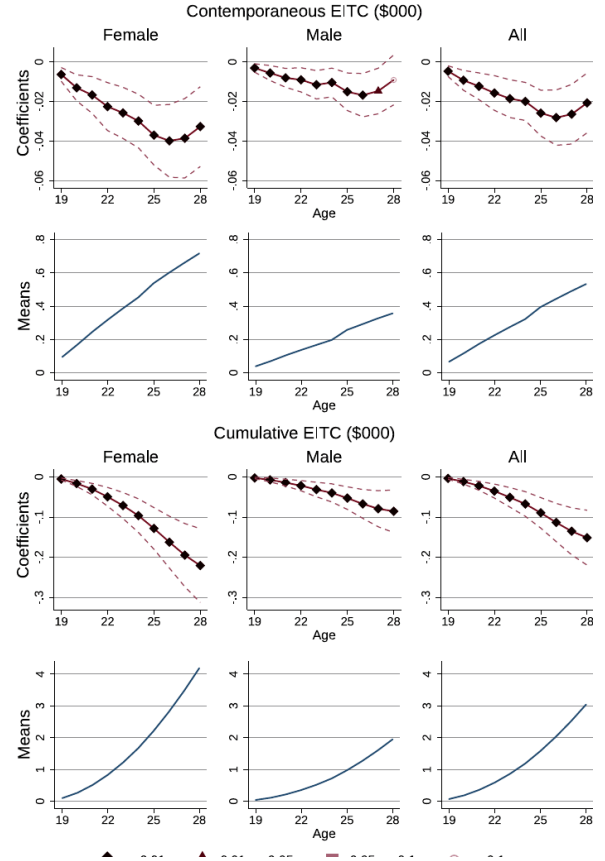


Figure 6: Contemporaneous and Cumulative EITC (\$000)

5.7 Figure 7: Contemporaneous and Cumulative Total Taxes (\$000)

Read:

- Medicaid exposure raises total tax payments, especially cumulatively and at later ages.
- By age 28, each additional simulated year of eligibility raises cumulative total taxes by about \$533 on a base of \$20,623.
- The tax effect combines higher income/payroll taxes and lower EITC payments.

Interpretation: Total taxes are the fiscal outcome that ties together human capital, fertility, and earnings. The estimates imply that childhood Medicaid partly pays for itself through higher adult tax payments.

Economic message: The fiscal return grows as beneficiaries age and labor-market outcomes mature. The by-age-28 fiscal return is likely not the full lifetime fiscal return.

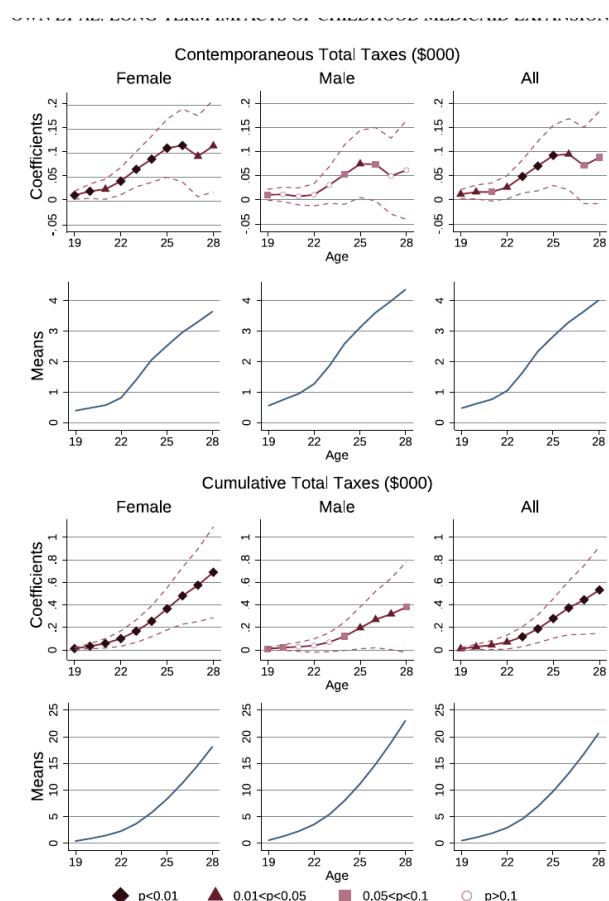


Figure 7: Contemporaneous and Cumulative Total Taxes (\$000)

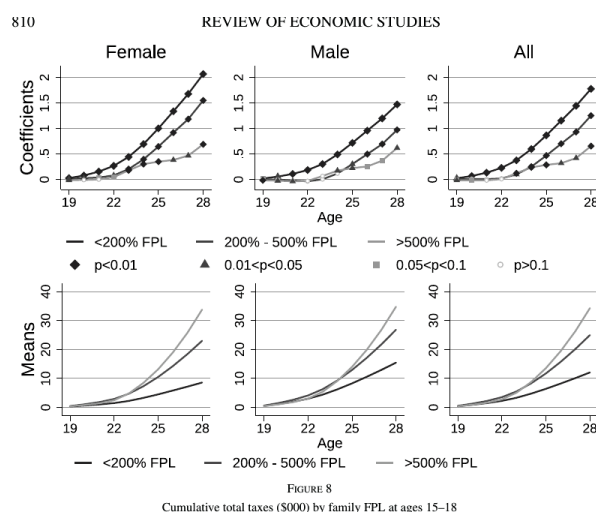


Figure 8: Cumulative Total Taxes (\$000) by Family FPL at Ages 15–18

Notes: Cumulative total taxes indicate taxes paid by a given age starting at age 19, defined as household federal tax payments plus individual payroll tax payments less EITC, adjusted to 2011 dollars. Coefficients for each age are obtained from separate reduced form regressions of total taxes on simulated years eligible, ages 0–18. Children are assigned to an % FPL bin if their household remained in that bin at every

5.8 Figure 8: Cumulative Total Taxes (\$000) by Family FPL at Ages 15–18

Read:

- The figure splits the sample into lower, middle, and higher FPL groups.

- Coefficients are largest for the below-200% FPL group, smaller for the 200–500% group, and smaller still for the above-500% group.
- At age 28, cumulative total taxes rise by about \$1,779 in the high-impact group, \$1,253 in the intermediate group, and \$656 in the low-impact group.

Interpretation: This dose-response pattern is one of the paper’s strongest identification checks. Medicaid should matter most for poor children, and the tax effects are largest in that group.

Economic message: The policy’s gains are concentrated among the children Medicaid was intended to help most, supporting both causal interpretation and distributional relevance.

5.9 Table 1: First Stage, Takeup, Spending, Taxes, and Fiscal ROI

Read:

- Column 1 shows a strong first stage for actual years eligible: one simulated year predicts 0.937 actual years eligible.
- Column 2 shows that one simulated year increases Medicaid uptake by 0.591 years.
- Column 3 shows that one simulated year raises Medicaid spending by \$593 without discounting and by \$404 at a 3% discount rate.
- Column 4 shows that one simulated year raises cumulative total taxes by \$533 without discounting and by \$233 at a 3% discount rate.
- Column 5 reports fiscal ROI. At a 3% discount rate, the government recoups about 58 cents per dollar by age 28, implying an ROI of about -42%.

Interpretation: The first stage validates the simulated eligibility measure. The fiscal ROI calculation shows that a sizable share of childhood Medicaid spending returns as adult tax revenue even before beneficiaries reach middle age.

Economic message: If benefits after age 28, private welfare gains, and mortality value are added, the social return is likely higher than the narrow fiscal ROI in Table 1.

5.10 Table A.1: Contemporaneous Outcomes at Ages 19, 22, and 28

Read:

- The table reports the age-specific contemporaneous estimates for the six main outcomes.
- It confirms that college and fertility effects are concentrated early, while wages and taxes become more important later.
- It makes cross-outcome comparisons easier than the full age-profile figures.

Interpretation: Table A.1 is useful for reading timing. It shows which outcomes respond at the same ages and helps connect early education/fertility changes to later labor-market and fiscal changes.

5.11 Table A.2: Cumulative Outcomes at Ages 19, 22, and 28

Read:

	(1) Years eligible for Medicaid, age 0–18	(2) Years of Medicaid takeup, age 0–18	(3) Medicaid spending (\$000), age 0–18	(4) Cumulative total taxes (\$000), age 19–28	(5) Fiscal ROI by age 28 = (3)/(2) – 100%
Discount rate = 0%					
Simulated years eligible, age 0–18	0.937***	0.591***	0.593***	0.533***	–10.20%
Mean	(0.077) 3.767	(0.037) 2.894	(0.065) 1.804	(0.192) 20.623	
Discount rate = 1%	–	–	0.520***	0.403***	–22.48%
Simulated years eligible, age 0–18			(0.057) 1.584	(0.145) 15.609	
Mean					
Discount rate = 2%	–	–	0.457***	0.306***	–33.11%
Simulated years eligible, age 0–18			(0.050) 1.396	(0.110) 11.846	
Mean					
Discount rate = 3%	–	–	0.404***	0.233***	–42.32%
Simulated years eligible, age 0–18			(0.045) 1.234	(0.837) 9.014	
Mean					
Observations	10,045,162	9,876,591	9,876,591	10,045,162	

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors in parentheses are clustered by state. Cumulative Medicaid take-up and spending for ages 0–18 are estimated in the full sample using administrative data from the MSIS, adjusted to 2011 dollars. Cumulative total taxes indicate taxes paid in the full sample from ages 19 to 28, defined as household federal tax payments plus individual payroll tax payments less EITC, adjusted to 2011 dollars. Coefficients are obtained from a reduced form regression of the given outcome on simulated years eligible, ages 0–18. The specification includes fixed effects for birth cohort by month and for state of residence at age 15 (the youngest age at which we observe all individuals in our sample). We exclude Arizona from the spending and take-up specifications due to missing MSIS data. The fiscal ROI by age 28 excluding Arizona from the cumulative total taxes specification is –8.23% (0% discount), –21.01% (1% discount), –31.72% (2% discount), and –41.17% (3% discount).

Table 1: First Stage, Takeup, Spending, Taxes, and Fiscal ROI

- The table reports cumulative effects through ages 19, 22, and 28.
- Cumulative college and fertility effects show early life-course shifts.
- Cumulative wage, EITC, and tax effects show the fiscal consequences that accumulate over early adulthood.

Interpretation: Table A.2 is the compact counterpart to the cumulative panels. It shows that cumulative fiscal effects are not one-year anomalies but build over time.

	(1) By age 19			(2) By age 22			(3) By age 28		
	Female	Male	All	Female	Male	All	Female	Male	All
College (currently enrolled, %)									
Simulated years eligible, age 0–18	1.261***	1.308**	1.281**	0.813**	0.806*	0.806*	0.458	0.519	0.486
Mean	(0.608) 62.209	(0.652) 51.593	(0.629) 56.785	(0.380) 73.562	(0.450) 62.858	(0.415) 68.094	(0.278) 80.888	(0.339) 69.501	(0.307) 75.070
Fertility (ever had dependent child, %)									
Simulated years eligible, age 0–18	–0.512**	–0.187	–0.348**	–0.870**	–0.387*	–0.625**	–1.177***	–0.721**	–0.948***
Mean	(0.212) 15.863	(0.119) 8.572	(0.159) 12.138	(0.344) 28.765	(0.215) 17.442	(0.271) 22.579	(0.313) 90.573	(0.352) 86.103	(0.332) 43.180
Mortality (%)									
Simulated years eligible, age 0–18	–0.003	–0.001	–0.002	–0.007*	0.007	–0.001	–0.009	–0.031*	–0.020*
Mean	(0.003) 0.037	(0.004) 0.008	(0.004) 0.008	(0.006) 0.152	(0.009) 0.454	(0.006) 0.306	(0.007) 0.417	(0.011) 1.191	(0.011) 0.812
Wage income (\$000)									
Simulated years eligible, age 0–18	0.059*	0.061	0.061*	0.152	–0.042	0.055	1.784***	0.581	1.177
Mean	(0.031) 4.198	(0.041) 4.864	(0.033) 4.538	(0.145) 25.262	(0.189) 30.147	(0.160) 27.758	(0.662) 136.600	(0.715) 161.350	(0.192) 149.245
EITC (\$000)									
Simulated years eligible, age 0–18	–0.006***	–0.003**	–0.003***	–0.029***	–0.026***	–0.042***	–0.263***	–0.103***	–0.180***
Mean	(0.002) 0.005	(0.001) 0.009	(0.001) 0.006	(0.015) 0.831	(0.008) 0.351	(0.011) 0.586	(0.063) 4.188	(0.034) 1.948	(0.006) 3.044
Total taxes (\$000)									
Simulated years eligible, age 0–18	0.013***	0.010*	0.012**	0.101***	0.099	0.099*	0.689***	0.380*	0.553***
Mean	(0.004) 0.391	(0.005) 0.555	(0.004) 0.475	(0.029) 2.263	(0.031) 2.363	(0.030) 2.913	(0.200) 18.115	(0.192) 23.025	(0.176) 20.623
Observations	49,133,139	5,132,023	10,045,162	49,133,139	5,132,023	10,045,162	49,133,139	5,132,023	10,045,162

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors in parentheses are clustered by state. Cumulative college enrollment indicates ever having enrolled in college by a given age, starting at age 19, observed through Form 1098-T, filed by educational institutions. Cumulative fertility indicates if a dependent child is ever born to a given age, starting at age 19. If an individual ever claims a dependent child on a Form 1040, SSA records yield age at birth. Cumulative mortality indicates mortality by a given age, starting at age 19, measured using SSA death records. Cumulative wage income indicates wage income earned by a given age, starting at age 19, obtained from Form W-2, adjusted to 2011 dollars and censored at \$10 million. Cumulative EITC indicates EITC earned by a given age, starting at age 19, obtained from Form 1040, adjusted to 2011 dollars. Cumulative total taxes indicate taxes paid by a given age, starting at age 19, defined as household federal tax payments plus individual payroll tax payments less EITC, adjusted to 2011 dollars. Coefficients for each age are obtained from separate reduced form regressions of the given outcome at that age on simulated years eligible, ages 0–18. The specification includes fixed effects for birth cohort by month and for state of residence at age 15 (the youngest age at which we observe all individuals in our sample).

Table A.2: Cumulative Outcomes at Ages 19, 22, and 28

	(1) By age 19			(2) By age 22			(3) By age 28		
	Female	Male	All	Female	Male	All	Female	Male	All
College (currently enrolled, %)									
Simulated years eligible, age 0–18	1.750***	1.637***	1.690***	0.718	0.646*	0.678*	–0.026	–0.092	–0.062
Mean	(0.539) 57.617	(0.568) 47.684	(0.549) 52.542	(0.435) 50.040	(0.342) 40.454	(0.384) 45.143	(0.157) 20.062	(0.154) 14.461	(0.143) 17.201
Fertility (first had dependent child, %)									
Simulated years eligible, age 0–18	–0.119**	–0.051	–0.085*	–0.088	–0.042	–0.065	–0.063	–0.072**	–0.068*
Mean	(–0.057) 4.848	(–0.042) 2.488	(0.047) 3.642	(0.054) 3.971	(0.048) 3.971	(0.045) 3.514	(0.050) 3.294	(0.029) 2.852	(0.034) 3.066
Mortality (%)									
Simulated years eligible, age 0–18	–0.001	–0.001	–0.002	–0.005**	0.006	0.000	0.006*	–0.009	–0.001
Mean	(0.003) 0.037	(0.004) 0.098	(0.003) 0.068	(0.002) 0.040	(0.003) 0.125	(0.003) 0.084	(0.003) 0.051	(0.006) 0.121	(0.004) 0.087
Wage income (\$000)									
Simulated years eligible, age 0–18	0.059*	0.061	0.061*	0.077	–0.045	0.015	0.414***	0.149	0.280*
Mean	(0.031) 4.198	(0.041) 4.864	(0.033) 4.538	(0.061) 8.729	(0.075) 10.461	(0.064) 9.614	(0.148) 23.356	(0.185) 28.377	(0.150) 26.013
EITC (\$000)									
Simulated years eligible, age 0–18	–0.006***	–0.003***	–0.005***	–0.023***	–0.009***	–0.016***	–0.033***	–0.009	–0.021***
Mean	(0.002) 0.005	(0.001) 0.009	(0.001) 0.006	(0.006) 0.319	(0.003) 0.137	(0.004) 0.226	(0.010) 0.717	(0.006) 0.357	(0.008) 0.533
Total taxes (\$000)									
Simulated years eligible, age 0–18	0.013***	0.010*	0.012**	0.042***	0.010	0.025*	0.115***	0.061	0.088*
Mean	(0.004) 0.391	(0.006) 0.555	(0.005) 0.475	(0.014) 0.815	(0.011) 1.271	(0.012) 1.068	(0.047) 3.640	(0.051) 4.864	(0.048) 4.010
Observations	49,133,139	5,132,023	10,045,162	49,133,139	5,132,023	10,045,162	49,133,139	5,132,023	10,045,162

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors in parentheses are clustered by state. Contemporaneous college enrollment indicates current enrollment in college at a given age, observed through Form 1098-T, filed by educational institutions. Contemporaneous fertility indicates if a first dependent child is born at a given age. If an individual ever claims a dependent child on a Form 1040, SSA records yield age at birth. Contemporaneous mortality indicates mortality at a given age, measured using SSA death records. Contemporaneous wage income indicates wages earned at a given age, obtained from Form W-2, adjusted to 2011 dollars and censored at \$10 million. Contemporaneous EITC indicates EITC earned at a given age, obtained from Form 1040, adjusted to 2011 dollars. Contemporaneous total taxes indicate taxes paid at a given age, defined as household federal tax payments plus individual payroll tax payments less EITC, adjusted to 2011 dollars. Coefficients for each age are obtained from separate reduced form regressions of the given outcome at that age on simulated years eligible, ages 0–18. The specification includes fixed effects for birth cohort by month and for state of residence at age 15 (the youngest age at which we observe all individuals in our sample).

Table A.1: Contemporaneous Outcomes at Ages 19, 22, and 28

	(1) By age 19			(2) By age 22			(3) By age 28		
	Female	Male	All	Female	Male	All	Female	Male	All
Main results									
Simulated years eligible, age 0–18	0.013***	0.010*	0.012**	0.101***	0.039	0.060**	0.689***	0.380*	0.553***
Mean	(0.004) 0.391	(0.006) 0.555	(0.005) 0.475	(0.014) 0.815	(0.011) 1.271	(0.012) 1.068	(0.047) 3.640	(0.051) 4.864	(0.048) 4.010
Instrument: simulated eligibility in the CPS									
Simulated years eligible, age 0–18	0.013***	0.012*	0.014**	0.114***	0.045	0.079**	0.731***	0.395*	0.560***
Mean	(0.005) 0.391	(0.006) 0.555	(0.005) 0.475	(0.009) 0.815	(0.033) 1.271	(0.035) 1.068	(0.035) 3.293	(0.216) 18.115	(0.202) 20.623
Sample: constant state of residence									
Simulated years eligible, age 0–18	0.013**	0.015*	0.014**	0.078**	0.040	0.059	0.376**	0.164	0.269
Mean	(0.006) 0.392	(0.008) 0.556	(0.007) 0.476	(0.038) 0.815	(0.035) 1.274	(0.036) 1.068	(0.036) 3.294	(0.180) 18.115	(0.167) 20.623
Specification: inclusion of sibling controls									
Simulated years eligible, age 0–18	–	–	0.013**	–	–	0.064*	–	–	0.312*
Mean	–	–	(0.005) 0.494	–	–	(0.034) 3.029	–	–	(0.172) 21.537
Specification: inclusion of family fixed effects									
Simulated years eligible, age 0–18	–	–	0.014*	–	–	0.052	–	–	0.035
Mean	–	–	(0.012) 0.494	–	–	(0.063) 3.029	–	–	(0.303) 21.537
Sample: parents filing jointly									
Simulated years eligible, age 0–18	0.009**	0.009	0.009*	0.065**	0.013	0.039	0.481**	0.222	0.350*
Mean	(0.004) 0.436	(0.006) 0.595	(0.005) 0.517	(0.011) 0.863	(0.030) 1.260	(0.030) 1.068	(0.200) 22.396	(0.226) 28.452	(0.207) 24.477
Sample: parents not filing jointly									
Simulated years eligible, age 0–18	0.020***	0.012**	0.016***	0.168***	0.084**	0.128***	0.854***	0.459***	0.651***
Mean	(0.006) 0.436	(0.005) 0.595	(0.005) 0.517	(0.044) 0.863	(0.035) 1.260	(0.035) 1.068	(0.229) 22.396	(0.163) 28.452	(0.183) 24.477
Sample: non-filers									
Simulated years eligible, age 0–18	0.013***	0.011**	0.012**	0.109***	0.038	0.074**	0.669***	0.322*	0.494***
Mean	(0.005) 0.369	(0.006) 0.538	(0.005) 0.455	(0.028) 0.815	(0.030) 1.271	(0.028) 1.068	(0.186) 21.706	(0.188) 23.356	(0.176) 20.623
Specification: inclusion of income controls									
Simulated years eligible, age 0–18	0.012***	0.010*	0.011**	0.082**	0.032	0.057*	0.451**	0.207	0.326*
Mean	(0.004) 0.391	(0.006) 0.555	(0.005) 0.475	(0.031) 0.815	(0.028) 1.271	(0.028) 1.068	(0.176) 21.706	(0.187) 23.356	(0.172) 20.623

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors in parentheses are clustered by state. Cumulative total taxes indicate taxes paid by a given age, starting at age 19, defined as household federal tax payments plus individual payroll tax payments less EITC, adjusted to 2011 dollars. Coefficients for each age are obtained from separate reduced form regressions of the given outcome at that age on simulated years eligible, ages 0–18. The specification includes fixed effects for birth cohort by month and for state of residence at age 15 (the youngest age at which we observe all individuals in our sample). The specifications that include sibling controls and family fixed effects are estimated in the sample of two-child

Table A.3: Robustness of Cumulative Total Tax (\$000) Results at Ages 19, 22, and 28

5.12 Table A.3: Robustness of Cumulative Total Tax (\$000) Results at Ages 19, 22, and 28

Read:

- The table reports robustness for cumulative total taxes under alternative simulation samples, sample restrictions, sibling controls, family fixed effects, filing-status splits, non-filer inclusion, and income controls.
- Most specifications preserve positive tax effects, especially by age 28.
- Income controls attenuate the estimate, which is expected if household income is partly affected by prior Medicaid exposure.

Interpretation: Table A.3 shows that the main tax result is not driven by one narrow specification. The weaker family fixed-effects estimates reflect limited within-family variation and possible household spillovers, not necessarily a contradiction of the main design.

6 Interpretive synthesis

- **Timing matters.** Medicaid effects do not appear all at once. Early adult effects are mainly education and fertility; later effects are wages and taxes.
- **Human capital and fertility are linked.** College enrollment rises when fertility falls, suggesting that Medicaid may shift early adult investments and family formation jointly.
- **Fiscal feedback is partial but large.** By age 28, tax revenue offsets a large fraction of spending. Since beneficiaries are still young, this is a lower-bound lifecycle fiscal perspective.
- **Mortality adds welfare value.** Narrow fiscal ROI understates social value because it excludes health and mortality benefits.
- **Distributional targeting matters.** The dose-response results show the strongest impacts for low-FPL children, consistent with Medicaid's targeting and with the identification logic.

7 Short summary

- The paper studies long-run impacts of childhood Medicaid expansions using IRS administrative tax records.
- The sample covers over 10 million children born between 1981 and 1984.
- The treatment is simulated years eligible for Medicaid from ages 0 to 18.
- The baseline specification includes birth-month cohort fixed effects and state-of-residence-at-age-15 fixed effects.
- Medicaid eligibility raises college enrollment, especially through age 22.
- Medicaid eligibility delays first dependent-child births, especially around ages 18 to 21.
- Cumulative mortality is lower by age 28.
- Female wage income rises starting around age 23; male wage effects are imprecise.
- EITC receipts fall, likely because of higher earnings and fertility changes.
- Total taxes rise, and cumulative tax effects grow with age.
- Table 1 shows that one simulated year of eligibility increases takeup by 0.591 years and spending by \$593.
- At a 3% discount rate, the government recoups 58 cents per dollar spent by age 28.

- Dose-response results are strongest for low-FPL children, supporting the Medicaid interpretation.
- Robustness exercises support the main tax result, though some within-family specifications are attenuated.

8 Conclusion

8.1 Core conclusion

Brown, Kowalski, and Lurie show that childhood Medicaid expansions have long-lasting effects on education, fertility, mortality, earnings, transfers, and tax payments. The policy raises human capital and fiscal outcomes in ways that are visible well into young adulthood.

8.2 Economic intuition

Medicaid changes childhood conditions through insurance coverage, health care access, financial protection, and family resources. These changes can improve schooling and delay fertility. Over time, better early-adult trajectories translate into higher earnings, lower transfer receipts, and higher tax payments.

8.3 Incremental contribution

The paper’s distinctive contribution is to connect childhood Medicaid exposure to a comprehensive set of administrative adult outcomes and to translate those outcomes into a fiscal return. It moves the literature from short-run coverage and utilization effects toward a lifecycle public-finance view of childhood health insurance.

8.4 Policy implication

The narrow fiscal ROI by age 28 is already substantial, and the broader social return is likely larger after accounting for later-life taxes, health gains, private welfare, and mortality reductions. The evidence supports evaluating childhood safety-net programs as investments rather than current-period transfers only.

8.5 Takeaway

Childhood Medicaid expansions generated persistent changes in adulthood. The paper demonstrates that public insurance for children can produce measurable long-run human-capital and fiscal returns, especially for low-income children.